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Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

OBJECTIVE 1:

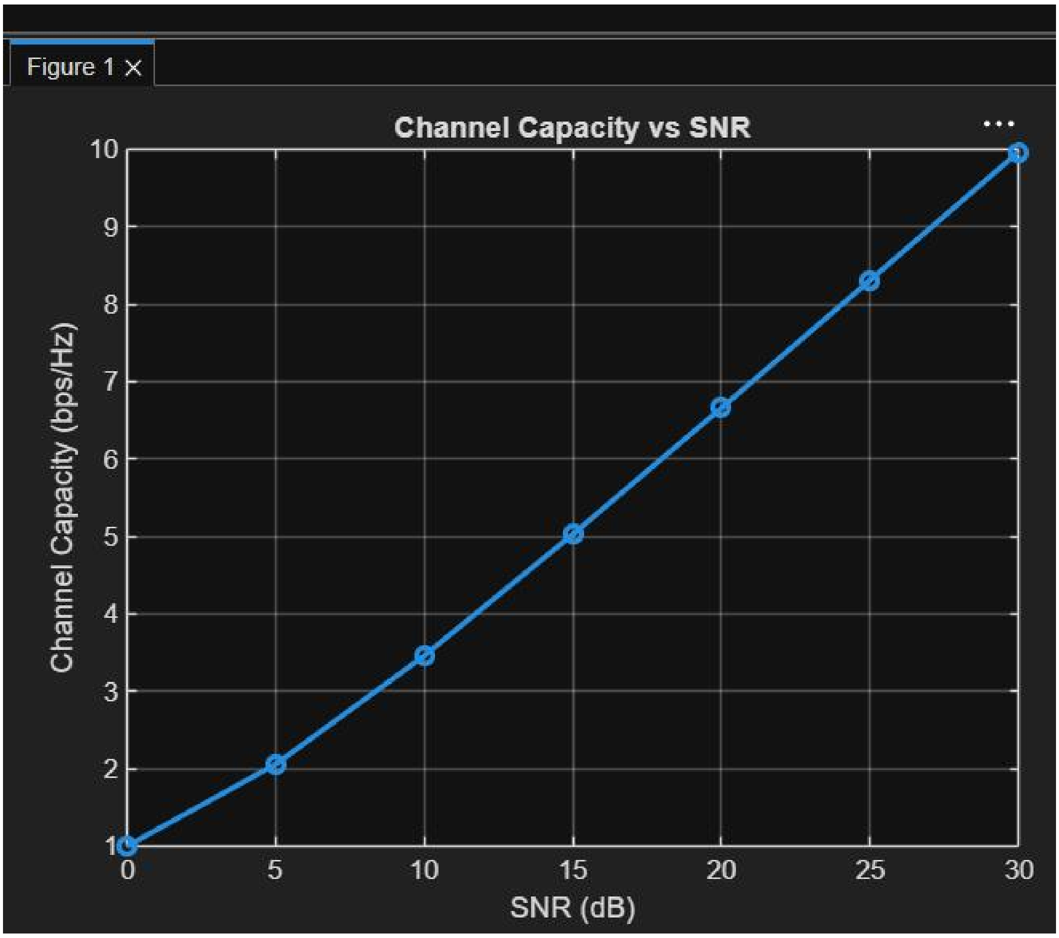
CODE:

```
clc;
clear;
close all;

snr = 0:5:30;                    % SNR values in dB
snr_linear = 10.^(snr/10);       % Convert dB to linear scale
capacity = log2(1 + snr_linear);  % Shannon Channel Capacity

figure;
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);
grid on;

xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



OBJECTIVE 2:

CODE:

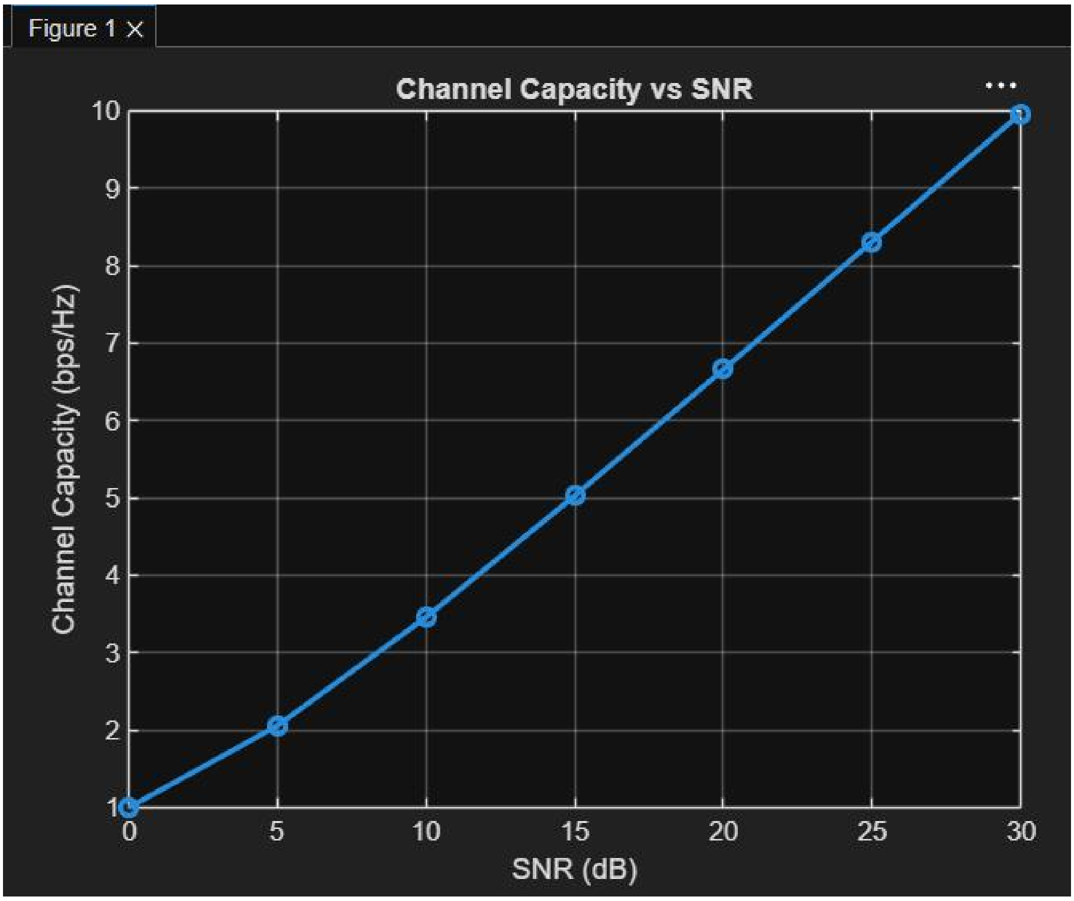
```
clc;
clear;
close all;

snr = 0:5:30;
snr_linear = 10.^(snr/10);        % Convert SNR from dB to linear scale
capacity = log2(1 + snr_linear); % Shannon Capacity

figure;
```

```
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);
grid on;

xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



OBJECTIVE 3:

CODE:

```
clc;
clear;
close all;
```

```
antennas = [4 8 16 32 64];     % Number of Antenna Elements
gain = 10*log10(antennas);     % Antenna Gain (dB)

figure
bar(antennas, gain)
grid on

xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```